

Tracing the Unseen: Trajectory Recovery from Cellular Records

(Online Appendix)

Jingyu Zhu [†], Yu Sun [†], Shaoxu Song [§], Dewang Ren [‡], Junhui Liu [‡], Xiaojie Yuan [†]

[†] Nankai University [§] Tsinghua University [‡] Xi'an Aeronautics Computing Technique Research Institute

{21202407094@mail., sunyu@, yuanxj@}nankai.edu.cn, sxsong@tsinghua.edu.cn, {rendewang, liujunhui}@avic.edu.cn

This document is the online appendix of the paper “*Tracing the Unseen: Trajectory Recovery from Cellular Records*”. It provides the complete proofs of all theoretical results. The numbering of propositions and lemmas follows the main paper, and each result is restated below before its proof.

Notation. $\mathcal{M}$ denotes the road network and $d_{\mathcal{M}}(\cdot, \cdot)$ the shortest-path (network) distance. $\mathcal{P}$ is the set of candidate locations, $\mathcal{R} \subseteq \mathcal{P}$ the set of representative candidates with $r = |\mathcal{R}|$, and n the number of timestamps $t_1, \dots, t_n$ of a cellular record sequence. $\Omega_i(p)$ is the coverage count of location p at timestamp t_i , i.e., the number of observed base stations whose coverage regions contain p , $\Omega(P) = \sum_{i=1}^n \Omega_i(p_i)$ is the overall coverage of a trajectory $P = \{p_1, \dots, p_n\}$, and $C_i^{\max} = \max_{p \in \mathcal{P}} \Omega_i(p)$. s is the speed constraint and δ the group diameter. TREC, TRRP, TRHA, and TRSN refer to Algorithms 1–4 of the main paper, respectively.

TREC is the exact dynamic programming algorithm based on the recurrence

$$\omega_{ij} = \begin{cases} \Omega_1(p_j) & i = 1 \\ \max_{\substack{p_k \in \mathcal{P} \\ d_{\mathcal{M}}(p_j, p_k) \leq s \cdot (t_i - t_{i-1})}} \{\omega_{i-1,k} + \Omega_i(p_j)\} & i > 1, \end{cases} \quad (1)$$

where ω_{ij} is the maximum cumulative coverage achievable from t_1 to t_i when the device is located at candidate p_j . TRHA greedily selects, at each timestamp t_i , a feasible candidate maximizing the score

$$S_i(p) = \alpha \cdot \frac{\Omega_i(p)}{C_i^{\max}} + (1 - \alpha) \cdot \left(1 - \frac{d_{\mathcal{M}}(p, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)}\right), \quad (2)$$

where $\bar{p}_{i+k}$ is the anchor selected within the k -step lookahead window and $\alpha \in (0, 1]$ balances coverage against movement consistency.

APPENDIX

A. Proof of Proposition 1

Proposition 1 (restated). *The exact trajectory recovery algorithm TREC returns the optimal trajectory P^* maximizing the overall coverage $\Omega(P)$ in $O(n|\mathcal{P}|^2)$ time, where n is the number of timestamps and $|\mathcal{P}|$ is the number of candidate locations.*

Proof: For correctness, TREC ensures optimal trajectory reconstruction by applying the dynamic programming recurrence in Equation 1. At each timestamp t_i , the state value ω_{ij} for candidate p_j is updated from feasible predecessors, i.e., candidates p_k with $d_{\mathcal{M}}(p_j, p_k) \leq s \cdot (t_i - t_{i-1})$, thereby preserving Bellman’s principle of optimality [1]. The predecessor index recorded at each update allows complete reconstruction by backtracking from the endpoint maximizing ω_n , guaranteeing that the recovered trajectory P^* maximizes $\Omega(P)$ under all constraints.

For complexity, the nested loops over n timestamps and all $|\mathcal{P}|^2$ candidate pairs yield $O(n \cdot |\mathcal{P}|^2)$. Each distance check $d_{\mathcal{M}}(p_h, p_j)$ can be performed in $O(1)$ time after a standard preprocessing of candidate-pair distances, a routine step also adopted in related studies [2], [3]. Thus the overall runtime of TREC is $O(n \cdot |\mathcal{P}|^2)$. ■

B. Proof of Lemma 2

Lemma 2 (Spatial Constraint) (restated). *Every group θ of candidate points constructed by the grouping procedure of TRRP satisfies the diameter constraint $\max_{p_i, p_j \in \theta} d_{\mathcal{M}}(p_i, p_j) < \delta$, where δ is the given group diameter.*

Proof: We prove the spatial constraint property by analyzing the region formation process of the grouping procedure in TRRP. Consider any region θ generated by the procedure. The region starts with a single point p_i , where the maximum pairwise distance is trivially $0 \leq \delta$. Points are then added to θ iteratively. The condition for adding a point p_j to θ requires that for all existing points $p_k \in \theta$, the distance $d_{\mathcal{M}}(p_j, p_k) < \delta$. This ensures that after each addition, the maximum pairwise distance within θ remains strictly less than δ . Therefore, for the final region θ , $\max_{p_i, p_j \in \theta} d_{\mathcal{M}}(p_i, p_j) < \delta$. ■

C. Proof of Proposition 3

Proposition 3 (restated). *TRRP returns the recovered trajectory P' in $O(|\mathcal{P}| \log |\mathcal{P}| + nr^2)$ time, where n is the number of timestamps and $r = |\mathcal{R}|$ is the number of representative candidates.*

Proof: TRRP first performs the grouping process. By using a kd-tree to identify neighbors within diameter δ , each

candidate is processed only once, yielding a complexity of $O(|\mathcal{P}| \log |\mathcal{P}|)$. Next, representatives are selected by computing the distance of each candidate to its group centroid, with total cost $O(|\mathcal{P}|)$. Finally, TREC is executed on the representative set with a complexity of $O(n \cdot r^2)$. Combining these stages, the overall complexity of the algorithm is $O(|\mathcal{P}| \log |\mathcal{P}| + n \cdot r^2)$. ■

D. Proof of Lemma 4

Lemma 4 (Bounded Coverage Variation) (restated). *Let L denote the maximum number of coverage-region boundaries crossed per unit length along any path in $\mathcal{M}$, determined by the density of base stations and the geometry of their coverage regions. Then for any two locations $p_j, p_k \in \mathcal{M}$, $|\Omega_i(p_j) - \Omega_i(p_k)| \leq L \cdot d_{\mathcal{M}}(p_j, p_k)$.*

Proof: The coverage count $\Omega_i(p)$ represents the number of base stations whose coverage regions contain location p . As a device moves from p_j to p_k along the road network, the coverage count changes only when a coverage-region boundary is crossed, gaining or losing connectivity to the corresponding base station.

The number of coverage-region boundaries crossed when moving a distance d along any path in $\mathcal{M}$ is bounded by the product of the distance and the maximum spatial density of coverage boundaries. Let $\rho_{\max}$ be the maximum density of coverage boundaries per unit length along any path in $\mathcal{M}$, determined by the density of base stations and the geometry of their coverage regions. Taking a shortest path between p_j and p_k , of length $d_{\mathcal{M}}(p_j, p_k)$, the change in coverage count satisfies $|\Omega_i(p_j) - \Omega_i(p_k)| \leq \rho_{\max} \cdot d_{\mathcal{M}}(p_j, p_k)$. Setting $L := \rho_{\max}$ yields the stated bound. ■

E. Proof of Proposition 5

Proposition 5 (Coverage Bound) (restated). *For the recovered approximate trajectory P' , the coverage satisfies $\Omega(P') \geq \Omega(P^*) - (2n + \gamma n^2)L\delta$, where n is the number of timestamps, L is the constant of Lemma 4, δ is the group diameter, and γ is a constant arising from restoring speed-constraint feasibility after grouping.*

Proof: Let $P^* = \{p_1^*, \dots, p_n^*\}$ denote the optimal trajectory over $\mathcal{P}$, and for each timestamp i let $\hat{p}_{\theta_i} \in \mathcal{R}$ denote the representative of the group θ_i containing p_i^* . Since p_i^* and $\hat{p}_{\theta_i}$ belong to the same group, Lemma 2 gives $d_{\mathcal{M}}(p_i^*, \hat{p}_{\theta_i}) < \delta$, and Lemma 4 yields the per-timestamp replacement loss

$$\Omega_i(p_i^*) - \Omega_i(\hat{p}_{\theta_i}) \leq L \cdot d_{\mathcal{M}}(p_i^*, \hat{p}_{\theta_i}) \leq L\delta. \quad (3)$$

The projected sequence $\hat{P} = \{\hat{p}_{\theta_1}, \dots, \hat{p}_{\theta_n}\}$, however, is not necessarily feasible: by the triangle inequality, $d_{\mathcal{M}}(\hat{p}_{\theta_i}, \hat{p}_{\theta_{i+1}}) \leq d_{\mathcal{M}}(p_i^*, p_{i+1}^*) + 2\delta$, so consecutive representatives may exceed the speed budget by up to 2δ . We therefore construct a feasible surrogate trajectory $Q = \{q_1, \dots, q_n\} \subseteq \mathcal{R}$ inductively: set $q_1 = \hat{p}_{\theta_1}$, and let q_{i+1} be the representative reachable from q_i under the speed constraint that is closest to $\hat{p}_{\theta_{i+1}}$. Because every candidate lies within distance δ of

its representative, each step of this construction increases the residual deviation $d_{\mathcal{M}}(q_i, \hat{p}_{\theta_i})$ by at most $c\delta$ for a small constant c : at most 3δ is lost when approaching $\hat{p}_{\theta_{i+1}}$ along a shortest path within the speed budget, and at most 2δ when snapping the resulting endpoint to its nearest representative. Hence $d_{\mathcal{M}}(q_i, \hat{p}_{\theta_i}) \leq c(i-1)\delta$. Combining both losses via Lemma 4 and the triangle inequality, and setting $\gamma := c/2$, we obtain

$$\Omega(P^*) - \Omega(Q) \leq \sum_{i=1}^n (\Omega_i(p_i^*) - \Omega_i(\hat{p}_{\theta_i})) + (\Omega_i(\hat{p}_{\theta_i}) - \Omega_i(q_i)) \quad (4)$$

$$\leq \sum_{i=1}^n (\delta + c(i-1)\delta) L \quad (5)$$

$$\leq (2n + \gamma n^2)L\delta. \quad (6)$$

Finally, Q is a feasible trajectory over the representative set $\mathcal{R}$, while TRRP returns the optimal trajectory P' over $\mathcal{R}$ by invoking TREC (Proposition 1). Therefore $\Omega(P') \geq \Omega(Q) \geq \Omega(P^*) - (2n + \gamma n^2)L\delta$. ■

F. Proof of Proposition 6

Proposition 6 (Approximation Guarantee) (restated). *TRRP returns an approximate trajectory P' such that $\Omega(P') \geq (1 - \epsilon)\Omega(P^*)$, where the group diameter is chosen as $\delta = \frac{\epsilon\Omega_{\max}}{(2n + \gamma n^2)L}$, in order to guarantee a $(1 - \epsilon)$ approximation.*

Proof: From Proposition 5, we have the coverage bound $\Omega(P') \geq \Omega(P^*) - (2n + \gamma n^2)L\delta$. Substituting the chosen grouping diameter $\delta = \frac{\epsilon\Omega_{\max}}{(2n + \gamma n^2)L}$ into the inequality yields $\Omega(P') \geq \Omega(P^*) - (2n + \gamma n^2)L \cdot \frac{\epsilon\Omega_{\max}}{(2n + \gamma n^2)L} = \Omega(P^*) - \epsilon\Omega_{\max}$. Since $\Omega(P^*) \geq \Omega_{\max}$, it follows that $\Omega(P') \geq \Omega(P^*) - \epsilon\Omega_{\max} \geq \Omega(P^*) - \epsilon\Omega(P^*) = (1 - \epsilon)\Omega(P^*)$. Thus, TRRP achieves a $(1 - \epsilon)$ approximation guarantee. ■

G. Proof of Proposition 7

Proposition 7 (restated). *TRHA returns the trajectory P' in $O(n|\mathcal{P}|)$ time after a one-off $O(|\mathcal{P}|^2)$ preprocessing of candidate-pair distances, where n is the number of timestamps and $|\mathcal{P}|$ is the number of candidate locations.*

Proof: First, the one-off preprocessing of candidate-pair distances requires $O(|\mathcal{P}|^2)$ time, after which each reachability check takes $O(1)$. With k -step lookahead, identifying the maximum-coverage candidates and selecting the anchor $\bar{p}_{i+k}$ both take $O(|\mathcal{P}|)$. For each candidate at t_i , computing the score in Equation 2 requires $O(|\mathcal{P}|)$ in total, and selecting the best candidate is subsumed in this bound. Thus each iteration costs $O(|\mathcal{P}|)$, leading to $O(n|\mathcal{P}|)$ across all timestamps. Together with the one-off preprocessing cost $O(|\mathcal{P}|^2)$, the total complexity is $O(|\mathcal{P}|^2 + n|\mathcal{P}|)$, i.e., $O(n|\mathcal{P}|)$ per trajectory after preprocessing. ■

H. Proof of Proposition 8

Proposition 8 (Coverage Bound of TRHA) (restated). *Given the trajectory $P' = \{p'_1, p'_2, \dots, p'_n\}$ obtained by TRHA with $\alpha \in (0, 1]$, and the optimal trajectory $P^* = \{p_1^*, p_2^*, \dots, p_n^*\}$, we have $\Omega(P') \geq \Omega(P^*) \cdot \min_{i \in [1, n]} \lambda_i - \frac{1-\alpha}{\alpha} n C_i^{\max}$, where $\lambda_i = \frac{\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p)}{\max_{q \in \mathcal{F}_i(P^*_{i-1})} \Omega_i(q)}$ is the ratio of the best coverage attainable at t_i within the feasible sets $\mathcal{F}_i(\cdot)$ of the approximate and optimal solutions given their prefixes, and $C_i^{\max} := \max_{i \in [1, n]} C_i^{\max}$.*

Proof: Let $\mathcal{F}_i(P'_{i-1})$ be the set of feasible candidates at timestamp t_i given the prefix $\{p'_1, \dots, p'_{i-1}\}$ of the approximate solution, and similarly $\mathcal{F}_i(P^*_{i-1})$ for the optimal solution. To measure the similarity between the feasible sets, we define the coverage ratio λ_i for each timestamp t_i as $\lambda_i = \frac{\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p)}{\max_{q \in \mathcal{F}_i(P^*_{i-1})} \Omega_i(q)}$.

Since $p_i^* \in \mathcal{F}_i(P^*_{i-1})$, we have $\max_{q \in \mathcal{F}_i(P^*_{i-1})} \Omega_i(q) \geq \Omega_i(p_i^*)$, and thus $\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p) \geq \lambda_i \cdot \Omega_i(p_i^*)$. The algorithm selects a point $p'_i \in \mathcal{F}_i(P'_{i-1})$ by maximizing the score function in Equation 2, so $S_i(p'_i) \geq S_i(p)$ for any $p \in \mathcal{F}_i(P'_{i-1})$. In particular, let p be the point achieving the maximum coverage in $\mathcal{F}_i(P'_{i-1})$. Provided the anchor $\bar{p}_{i+k}$ is reachable from p within the lookahead window, i.e., $d_{\mathcal{M}}(p, \bar{p}_{i+k}) \leq s \cdot (t_{i+k} - t_i)$, the movement-consistency term of $S_i(p)$ is nonnegative and may be dropped, yielding:

$$\alpha \cdot \frac{\Omega_i(p'_i)}{C_i^{\max}} + (1 - \alpha) \cdot \left(1 - \frac{d_{\mathcal{M}}(p'_i, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)}\right) \geq \alpha \cdot \frac{\Omega_i(p)}{C_i^{\max}}. \quad (7)$$

Rearranging and combining with the earlier inequality for $\max_{p \in \mathcal{F}_i(P'_{i-1})} \Omega_i(p)$, we get

$$\Omega_i(p'_i) \geq \lambda_i \cdot \Omega_i(p_i^*) - \frac{(1 - \alpha)}{\alpha} C_i^{\max} \cdot \left(1 - \frac{d_{\mathcal{M}}(p'_i, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)}\right). \quad (8)$$

Each parenthesized term is at most 1. For the last k timestamps no anchor is applied, so the score reduces to the coverage term and $\Omega_i(p'_i) \geq \lambda_i \cdot \Omega_i(p_i^*)$ holds directly without penalty. Summing over $i = 1$ to n therefore yields

$$\Omega(P') = \sum_{i=1}^n \Omega_i(p'_i) \quad (9)$$

$$\geq \sum_{i=1}^n \lambda_i \cdot \Omega_i(p_i^*) - \frac{(1 - \alpha)}{\alpha} \sum_{i=1}^{n-k} C_i^{\max} \cdot \left(1 - \frac{d_{\mathcal{M}}(p'_i, \bar{p}_{i+k})}{s \cdot (t_{i+k} - t_i)}\right) \quad (10)$$

$$\geq \Omega(P^*) \cdot \min_{i \in [1, n]} \lambda_i - \frac{(1 - \alpha)}{\alpha} n C_i^{\max}, \quad (11)$$

where $C_i^{\max} = \max_{i \in [1, n]} C_i^{\max}$. ■

I. Proof of Proposition 9

Proposition 9 (restated). *Replacing $\mathcal{M}$ with the localized subnetwork $\mathcal{M}'$ preserves the optimal trajectory. Specifically, solving the recovery problem on $\mathcal{M}'$ yields a trajectory P'*

satisfying the speed constraint with $\Omega(P') = \Omega(P^)$, where P^* is the optimal solution on $\mathcal{M}$.*

Proof: Coverage is preserved since each $p_i \in D_c$ for some $c \in C_i$ belongs to $U \subseteq \mathcal{M}'$, so $\Omega_i(p_i)$ is identical in $\mathcal{M}$ and $\mathcal{M}'$. Feasibility is preserved because for any consecutive points p_i, p_{i+1} , a shortest path π in $\mathcal{M}$ can be decomposed into segments either inside R or outside with endpoints on ∂R . Internal segments are included in $\mathcal{M}'$, and for external segments, shortest paths between boundary points are preserved. Replacing each segment by its preserved path yields a path in $\mathcal{M}'$ of the same length, so $d_{\mathcal{M}'}(p_i, p_{i+1}) = d_{\mathcal{M}}(p_i, p_{i+1})$ and the speed constraint is maintained.

Since both coverage and feasibility are preserved, the dynamic programming recurrence is identical under $\mathcal{M}$ and $\mathcal{M}'$, and thus Algorithm TREC produces the same optimal trajectory in both networks. For Algorithm TRRP, as grouping is performed entirely within the subnetwork $\mathcal{M}'$, the reconstructed trajectory on $\mathcal{M}'$ is identical to what would be obtained on the original network. For TRHA, the heuristic relies on selecting future high-coverage points within a lookahead window; such points necessarily lie within $\mathcal{M}'$, or selecting alternatives outside $\mathcal{M}'$ cannot yield higher coverage. Similarly, the weighted selection of maximum-coverage candidates remains valid within $\mathcal{M}'$, ensuring that the approximate trajectory generated by TRHA on the subnetwork achieves equivalent coverage behavior as if computed on the full network. ■

J. Proof of Proposition 10

Proposition 10 (restated). *After applying TRSN optimization, the time complexity becomes $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} n \eta_{\theta}^2 |\mathcal{P}|^2)$ for TREC; $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} (\eta_{\theta}^2 |\mathcal{P}|^2 + n \eta_{\theta}^2 r^2))$ for TRRP; and $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} n \eta_{\theta} |\mathcal{P}|)$ for TRHA, where $|\Theta|$ is the number of subnetworks, M_{θ} is the number of sequences in subnetwork θ , n is the maximum number of timestamps among the sequences in θ , $|\mathcal{P}|$ is the original candidate set size, r is the number of representatives, and η_{θ} (with $0 < \eta_{\theta} < 1$) is the reduction ratio per subnetwork.*

Proof: The time complexity analysis follows from applying the TRSN optimization to each algorithm's base complexity. The key insight is that by partitioning the problem into independent subnetworks, we can analyze the complexity per subnetwork and then sum over all subnetworks.

For TREC, the base complexity is $O(n|\mathcal{P}|^2)$. After partitioning, for each subnetwork θ with M_{θ} cellular record sequences and candidate set size reduced to $\eta_{\theta}|\mathcal{P}|$ due to spatial localization, the complexity becomes $O(M_{\theta} n \eta_{\theta}^2 |\mathcal{P}|^2)$. Summing over all subnetworks gives $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} n \eta_{\theta}^2 |\mathcal{P}|^2)$. For TRRP, after partitioning, the grouping complexity per subnetwork is $O(\eta_{\theta}^2 |\mathcal{P}|^2)$ and the next recovery complexity is $O(n \eta_{\theta}^2 r^2)$. Applied to M_{θ} sequences, the total complexity becomes $O(\sum_{\theta=1}^{|\Theta|} M_{\theta} (\eta_{\theta}^2 |\mathcal{P}|^2 + n \eta_{\theta}^2 r^2))$. For TRHA, the base algorithm has complexity $O(n|\mathcal{P}|)$. After partitioning, the

per subnetwork complexity is $O(n\eta_\theta|\mathcal{P}|)$ for M_θ sequences, giving total complexity $O(\sum_{\theta=1}^{|\Theta|} M_\theta n\eta_\theta|\mathcal{P}|)$.

The reduction ratios η_θ , $0 < \eta_\theta < 1$ capture the spatial localization effect, representing the fraction of the original candidate set that remains relevant within each subnetwork θ . ■

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